Dynamical Systems Meets Data: DMD, Koopman, and Reduced Representations

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AI4PEX

Dynamical Systems Meets Data

Part 1: DMD, Koopman, and Reduced Representations

- Dynamical systems basics, attractors, chaos
- Meeting data & Koopman operators
- Dynamic mode decomposition & algorithms

Part 2: Nonlinearity, Stochasticity, and Forcing

Please do ask questions!

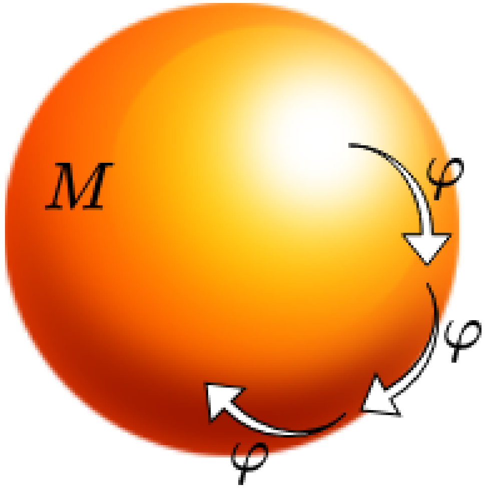
What is a dynamical system?

Dynamical System

Dynamical system is a **mathematical object** defined by a state space M and an operator φ_τ defining evolution of any initial state in M over time $\tau > 0$:

- state space M
- evolution operator $\varphi_\tau : M \to M$

  $\mathbf{x}(t_0 + \tau) = \varphi_\tau(\mathbf{x}(t_0))$ for any $\mathbf{x}(t_0) \in M$ and $\tau > 0$
- cocycle property $\varphi_{\tau_1+\tau_2} = \varphi_{\tau_2} \circ \varphi_{\tau_1}$



Dynamical systems provide a universal language to bridge models and real world

Dynamical System: example

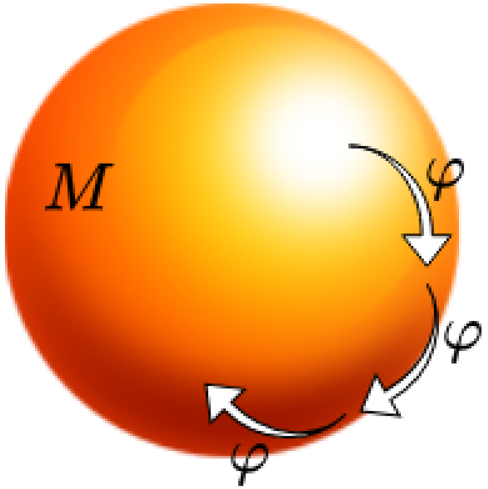
1st order ordinary differential equation (ODE) / flow:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

- state space $M = \mathbb{R}^d$
- evolution operator over time τ:

$$\mathbf{x}(t_0 + \tau) = \mathbf{x}(t_0) + \int_{t_0}^{t_0+\tau} \mathbf{f}(\mathbf{x}(t'))dt'$$

- cocycle property $\varphi_{\tau_1+\tau_2} = \varphi_{\tau_2} \circ \varphi_{\tau_1}$ (check!)



Dynamical System: example

1st order ordinary differential equation (ODE) / flow:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

- continuous or discrete time?

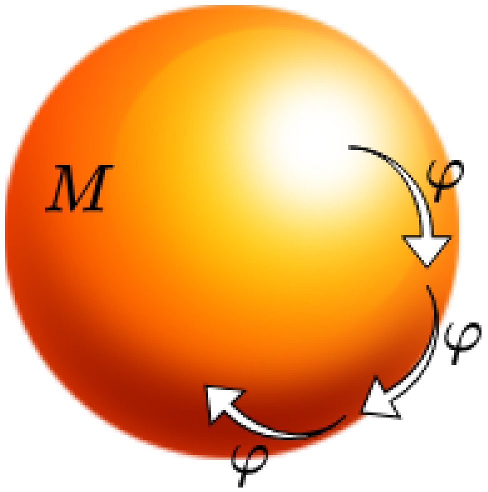
Dynamical System: discrete time

A 1-step map:

$$\mathbf{x}_{t+1} = \mathbf{F}(\mathbf{x}_t)$$

- state space $M = \mathbb{R}^d$
- evolution operator over τ steps:

  $\mathbf{x}_{t+\tau} = \mathbf{F}(\ldots(\mathbf{F}(\mathbf{x}_t))) = \mathbf{F}^{(\tau)}(\mathbf{x}_t)$
- cocycle property $\varphi_{\tau_1+\tau_2} = \varphi_{\tau_2} \circ \varphi_{\tau_1}$ (check!)



Dynamical System: discrete time

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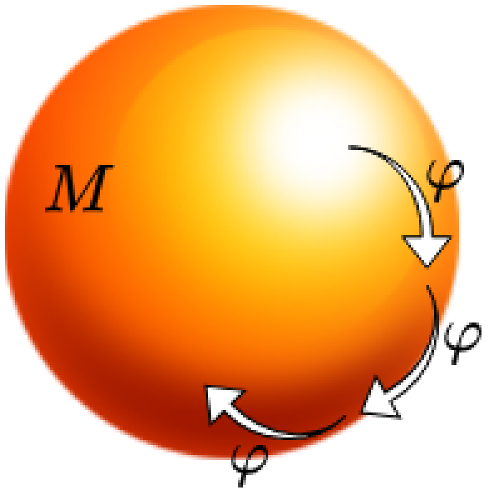

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- cocycle property $\varphi_{\tau_1+\tau_2} = \varphi_{\tau_2} \circ \varphi_{\tau_1}$ (check!)

Example:

Define 1-step $\mathbf{F}$ as the result of ODE forward integration over time step Δt

$$\mathbf{F}(\mathbf{x}(t_0)) := \mathbf{x}(t_0) + \int_{t_0}^{t_0+\Delta t} \mathbf{f}(\mathbf{x}(t'))dt'$$

Dynamical System: example

1st order ordinary differential equation (ODE) / flow:

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- **continuous** or discrete time?
- deterministic or stochastic?

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- autonomous or non-autonomous?

Dynamical System: example

1st order ordinary differential equation (ODE) / flow:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

- **continuous** or discrete time?
- **deterministic** or stochastic?
- **autonomous** or non-autonomous?
- linear or nonlinear?

Linear Dynamical System (1D)

Scalar ODE:

$$\dot{x} = \lambda x, \quad \lambda \in \mathbb{R}$$

- solution: $x(t) = e^{\lambda(t-t_0)} x(t_0)$
- evolution operator: $\varphi_\tau(x) = e^{\lambda\tau} x$
- behavior determined by sign of λ:
  - $\lambda < 0$: decay to zero
  - $\lambda = 0$: constant
  - $\lambda > 0$: exponential growth

Linear Dynamical System (1D)

Scalar map:

$$x_{t+1} = \mu x_t, \quad \mu \in \mathbb{R}$$

- solution: $x_t = \mu^{t-t_0} x_{t_0}$
- evolution operator: $\varphi_\tau(x) = \mu^\tau x$
- behavior determined by $|\mu|$:
  - $|\mu| < 1$: decay
  - $|\mu| = 1$: constant
  - $|\mu| > 1$: growth
- Link to continuous ODE discretized with the step Δt: $\mu = e^{\lambda \Delta t}$

Linear Dynamical System (2D)

Example: classic linear oscillator (q — position, v — velocity):

$$\dot{q} = v$$
$$\dot{v} = -\omega^2 q - 2\gamma v$$

- ω — frequency
- $0 \leq \gamma < \omega$ — damping (weak dissipation)
- state vector $\mathbf{x} = (q, v)^\top \in \mathbb{R}^2$

Linear Dynamical System (2D)

Example: classic linear oscillator (q — position, v — velocity):

$$\dot{q} = v$$
$$\dot{v} = -\omega^2 q - 2\gamma v$$

- solution ($\gamma < \omega$):

$$q(t) = e^{-\gamma(t-t_0)} \left[q(t_0) \cos \omega_d (t - t_0) + \frac{v(t_0) + \gamma q(t_0)}{\omega_d} \sin \omega_d (t - t_0) \right]$$

$$v(t) = e^{-\gamma(t-t_0)} \left[(v(t_0) + \gamma q(t_0)) \cos \omega_d (t - t_0) - \frac{\omega^2 q(t_0) + \gamma v(t_0)}{\omega_d} \sin \omega_d (t - t_0) \right]$$

- amplitude decays as $e^{-\gamma(t-t_0)}$
- oscillates at damped frequency $\omega_d = \sqrt{\omega^2 - \gamma^2} < \omega$

Linear Dynamical System (2D) — complex form

$$\dot{\mathbf{x}} = A\mathbf{x}, \qquad A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -2\gamma \end{pmatrix}$$

A: eigenvalues $\lambda_{1,2} = -\gamma \pm i\omega_d$, eigenvectors $\mathbf{v}_{1,2}$ (right), $\mathbf{w}_{1,2}$ (left)

In the eigenbasis $z_{1,2} = \mathbf{w}_{1,2}^\top \mathbf{x}$:

$$\dot{z}_{1,2} = \lambda_{1,2} z_{1,2}$$

Solution:

$$z_{1,2}(t) = e^{\lambda_{1,2}(t-t_0)} z_{1,2}(t_0) = e^{(-\gamma \pm i\omega_d)(t-t_0)} z_{1,2}(t_0)$$

- $e^{\pm i\omega_d(t-t_0)}$ — **rotation** in the complex plane with angular frequency $\pm\omega_d$
- $e^{-\gamma(t-t_0)}$ — **scaling** (decay)

Complex eigenvalues $\Leftrightarrow$ oscillatory (rotating) behavior in 2D

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Switch to original basis:

$$\mathbf{x}(t) = z_1(t)\,\mathbf{v}_1 + z_2(t)\,\mathbf{v}_2 = 2\,e^{-\gamma(t-t_0)}\,\mathrm{Re}\Big[e^{i\omega_d(t-t_0)} z_1(t_0)\,\mathbf{v}_1\Big]$$

Trajectory

- A **trajectory** (orbit) is a curve $t \mapsto \mathbf{x}(t) \in M$ satisfying:

$$\mathbf{x}(t_2) = \varphi_{t_2 - t_1}(\mathbf{x}(t_1)), \qquad t_2 \geq t_1$$

- The **phase portrait** is the collection of all trajectories in M. It reveals the global geometry of the dynamical system
- For a continuous dynamical system
  - for any $\mathbf{x}^* \in M$ there exists exactly one trajectory passing through $\mathbf{x}^*$, and distinct trajectories **never intersect**
  - $\mathbf{f}(\mathbf{x})$ is the velocity along the trajectory

Fixed point (equilibrium)

A **fixed point** (equilibrium) is a special trajectory $\mathbf{x}(t) = \mathbf{x}^*$ for all t, defined by:

$$\mathbf{f}(\mathbf{x}^*) = \mathbf{0} \quad (\text{continuous}), \qquad \mathbf{F}(\mathbf{x}^*) = \mathbf{x}^* \quad (\text{discrete})$$

Examples:

- **Linear 1D** — $\dot{x} = \lambda x$: unique equilibrium $x^* = 0$ for any λ
- **Linear 2D** — $\dot{\mathbf{x}} = A\mathbf{x}$: unique equilibrium $\mathbf{x}^* = \mathbf{0}$ if $\det(A) \neq 0$

Stability of fixed point

- A fixed point $\mathbf{x}^*$ is **stable** if trajectories starting close to $\mathbf{x}^*$ converge to it:

$$\|\mathbf{x}(t_0) - \mathbf{x}^*\| < \delta \implies \|\mathbf{x}(t) - \mathbf{x}^*\| \to 0 \quad \text{as } t \to \infty$$

- A fixed point is **unstable** if arbitrarily small perturbations lead to large deviations from $\mathbf{x}^*$.
- A fixed point is **non-hyperbolic** if some trajectories starting near $\mathbf{x}^*$ neither converge to it nor escape from it. It is structurally unstable under small perturbations to the system

Linear Dynamical System (2D)

Matrix form:

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad A \in \mathbb{R}^{2\times 2}, \quad \mathbf{x} \in \mathbb{R}^2$$

- solution: $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$
- behavior governed by eigenvalues $\lambda_{1,2}$ of A

Example — classic linear oscillator:

$$A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -2\gamma \end{pmatrix}$$

- $\gamma = 0$: pure oscillation (non-hyperbolic), eigenvalues $\lambda = \pm i\omega$
- $\gamma > 0$: damped oscillation (stable), $\mathrm{Re}(\lambda) < 0$

Linear Dynamical System (2D): eigenvalue cases

Behavior of $\dot{\mathbf{x}} = A\mathbf{x}$ determined by eigenvalues $\lambda_{1,2}$:

Complex eigenvalues $\lambda = \alpha \pm i\beta$, $\beta \neq 0$:

- $\alpha < 0$: **stable focus** — spiral inward
- $\alpha = 0$: **center** — closed orbits
- $\alpha > 0$: **unstable focus** — spiral outward

Real eigenvalues $\lambda_1, \lambda_2 \in \mathbb{R}$:

- $\lambda_2 < \lambda_1 < 0$: **stable node** — decay along both eigenvectors
- $\lambda_2 < 0 < \lambda_1$: **saddle** — attract along one, repel along other
- $0 < \lambda_2 < \lambda_1$: **unstable node** — growth along both eigenvectors

Key quantity: $\mathrm{Re}(\lambda_i) < 0$ for all $i \Leftrightarrow$ **stable** (all trajectories decay to origin)

Stability of a trajectory: Lyapunov exponents

Consider a perturbation $\boldsymbol{\delta}(t_0)$ to initial condition $\mathbf{x}(t_0)$ of a trajectory $\mathbf{x}(t)$.

- In a linear system perturbation satisfies $\dot{\boldsymbol{\delta}} = A\boldsymbol{\delta}$, so:

$$\boldsymbol{\delta}(t) = e^{A(t-t_0)}\boldsymbol{\delta}(t_0) = \sum_i c_i\, e^{\lambda_i(t-t_0)}\mathbf{v}_i, \qquad c_i = \mathbf{w}_i^\top \boldsymbol{\delta}(t_0)$$

- Each component grows or decays independently with rate $\sigma_i = \mathrm{Re}(\lambda_i)$.

This motivates the definition: the i-th **Lyapunov exponent** is the asymptotic growth rate of $\boldsymbol{\delta}$ along the i-th direction $\mathbf{v}_i$:

$$\sigma_i = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\boldsymbol{\delta}_i(t)\|}{\|\boldsymbol{\delta}_i(0)\|}$$

- The trajectory is stable if $\sigma_{\max} < 0$ (predictability!)

Linear dynamical system properties

- Linear system has the only fixed point (if $\det(A) \neq 0$)
- Dynamics is fully determined by eigenvalues of A
- General solution is a superposition of modes $e^{\lambda_i t}\mathbf{v}_i$, each grows or decays independently
- No oscillatory attractors, no chaos

Nonlinear dynamical systems, attractors, chaos

Nonlinear dynamical system (Lorenz-63)

Atmospheric convection toy model:

$$\begin{aligned}\dot{x} &= \sigma(y - x)\\ \dot{y} &= x(\rho - z) - y\\ \dot{z} &= xy - \beta z\end{aligned}$$

- State space $M = \mathbb{R}^3$
- Dynamics depends on the parameters ($\sigma = 10,\ \beta = 8/3,\ \rho > 0$)

Nonlinear dynamical system (Lorenz-63)

$$\begin{aligned}
\dot{x} &= \sigma(y - x) \\
\dot{y} &= x(\rho - z) - y \\
\dot{z} &= xy - \beta z
\end{aligned}$$

How to find fixed points (equilibrium states)?

Nonlinear dynamical system (Lorenz-63)

$$\dot{x} = \sigma(y - x)$$
$$\dot{y} = x(\rho - z) - y$$
$$\dot{z} = xy - \beta z$$

How to find fixed points (equilibrium states)?

Fixed points ($\mathbf{f}(\mathbf{x}^*) = \mathbf{0}$):

- (0, 0, 0)
- $C^{\pm} = \left(\pm\sqrt{\beta(\rho - 1)},\ \pm\sqrt{\beta(\rho - 1)},\ \rho - 1\right)$, if $\rho > 1$

Stability of fixed points

Linearization around fixed point:

- Given a nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ near an equilibrium $\mathbf{x}^*$, let $\boldsymbol{\xi} = \mathbf{x} - \mathbf{x}^*$:

$$\dot{\boldsymbol{\xi}} = D\mathbf{f}(\mathbf{x}^*)\,\boldsymbol{\xi} + \mathcal{O}(\|\boldsymbol{\xi}\|^2)$$

- **Jacobian** $J = D\mathbf{f}(\mathbf{x}^*)$ governs local dynamics
- valid for infinitely small perturbations $\boldsymbol{\xi}$

Hartman–Grobman theorem: if J has no eigenvalues with $\mathrm{Re}(\lambda) = 0$, the nonlinear flow is topologically equivalent to the linear flow $\dot{\boldsymbol{\xi}} = J\boldsymbol{\xi}$ near $\mathbf{x}^*$

Stability is determined by the eigenvalues of Jacobian

Nonlinear dynamical system (Lorenz-63)

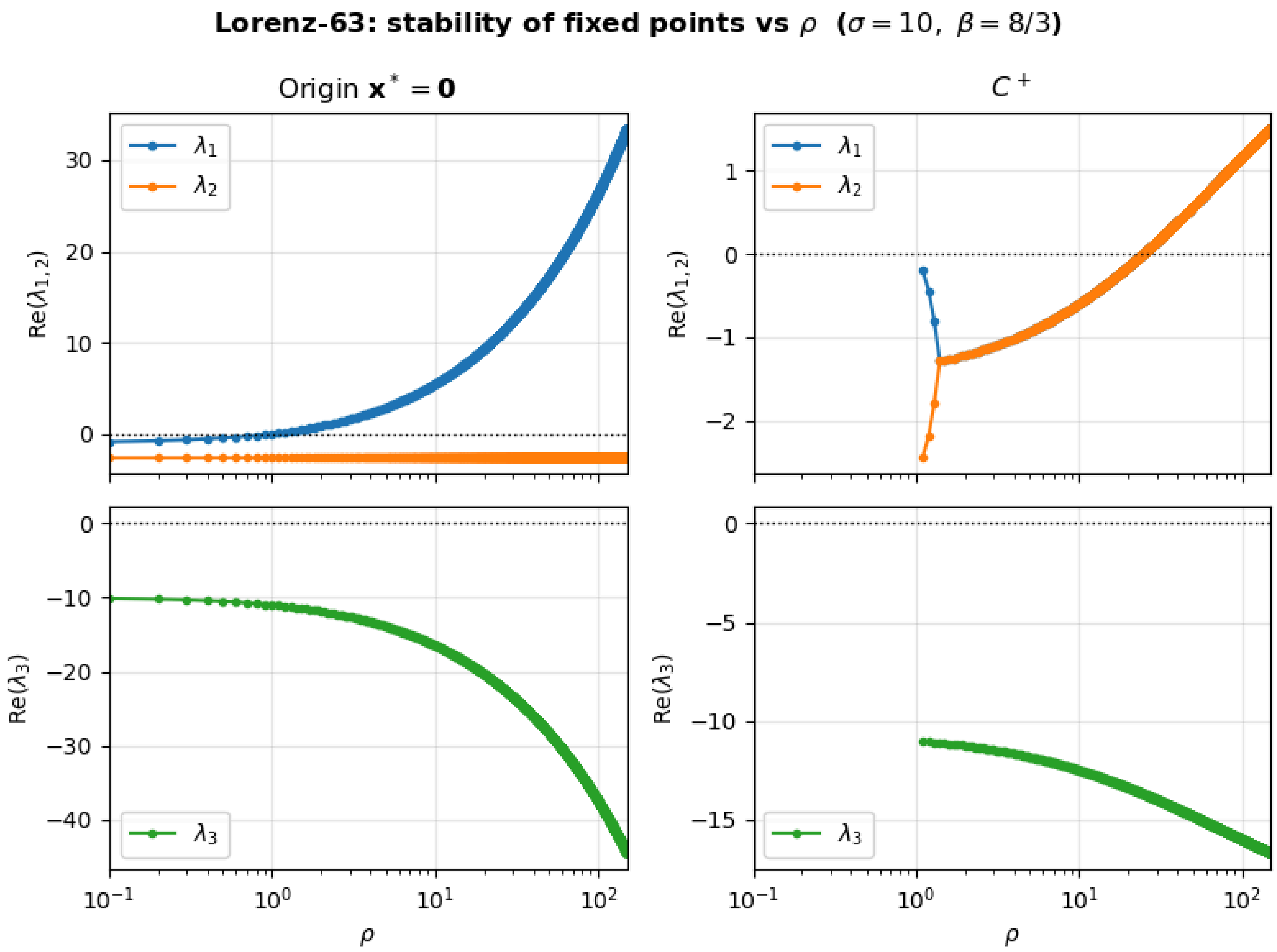
$$\dot{x} = \sigma(y - x)$$
$$\dot{y} = x(\rho - z) - y$$
$$\dot{z} = xy - \beta z$$

The Jacobian of $\mathbf{f}$ at any point (x, y, z):

$$J = D\mathbf{f} = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix}, \qquad \det(J) = -\sigma\beta(z - \rho + 1) - \sigma x(x + y)$$

- At $\mathbf{x}^* = \mathbf{0}$: $\det(J_0) = \sigma\beta(\rho - 1) > 0$ -- unstable for $\rho > 1$
- At $C^{\pm}$ for $\rho > 1$: $\det(J_{C^{\pm}}) = -2\sigma\beta(\rho - 1) < 0$ -- can be stable
- github.com/IPL-UV/dynamical_systems_course

Nonlinear dynamical system (Lorenz-63)



Nonlinear dynamical system (Lorenz-63)

$$\begin{aligned}\dot{x} &= \sigma(y - x)\\ \dot{y} &= x(\rho - z) - y\\ \dot{z} &= xy - \beta z\end{aligned}$$

Where do the trajectories go if all equilibrium points are unstable?

The system is globally stable:

There exists an ellipsoid $ax^2 + by^2 + cz^2 = r^2$ which traps all incoming trajectories (all $r > r_0$)

Attractor

A set $\mathcal{A} \subset M$ is an **attractor** if:

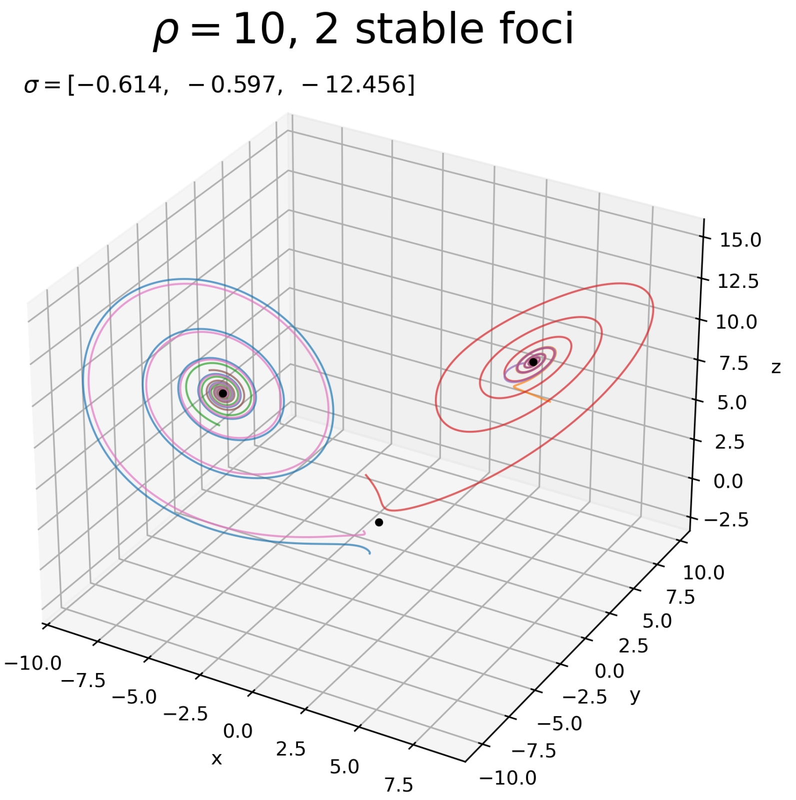
1. $\mathcal{A}$ is forward invariant: $\varphi_t(\mathcal{A}) \subseteq \mathcal{A}$ for all $t > 0$
2. There exists an open neighborhood $U \supset \mathcal{A}$ such that:

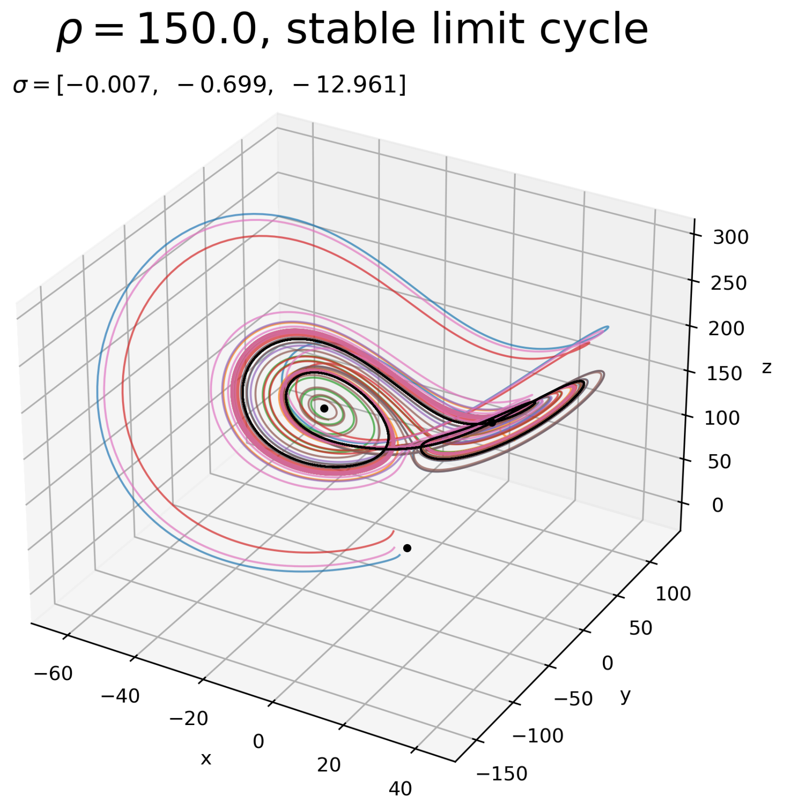
$$\varphi_t(\mathbf{x}) \to \mathcal{A} \quad \textbf{as } t \to \infty \quad \forall \mathbf{x} \in U$$

3. There is no non-empty subset of $\mathcal{A}$ satisfying both properties above

The **basin of attraction** $B(\mathcal{A})$ is the set of all $\mathbf{x} \in M$ whose trajectories converge to $\mathcal{A}$.

Nonlinear dynamical system (Lorenz-63)





Stability of a trajectory: Lyapunov exponents

Measure the **average rate of divergence** of nearby trajectories.

For a trajectory $\mathbf{x}(t)$ and perturbation $\boldsymbol{\delta}(t)$:

$$\dot{\boldsymbol{\delta}} = D\mathbf{f}(\mathbf{x}(t))\,\boldsymbol{\delta}$$

The i-th Lyapunov exponent:

$$\sigma_i = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\boldsymbol{\delta}_i(t)\|}{\|\boldsymbol{\delta}_i(0)\|}$$

Lyapunov exponents of an attractor

Measure the **average rate of divergence** of nearby trajectories.

For a trajectory $\mathbf{x}(t)$ and perturbation $\boldsymbol{\delta}(t)$:

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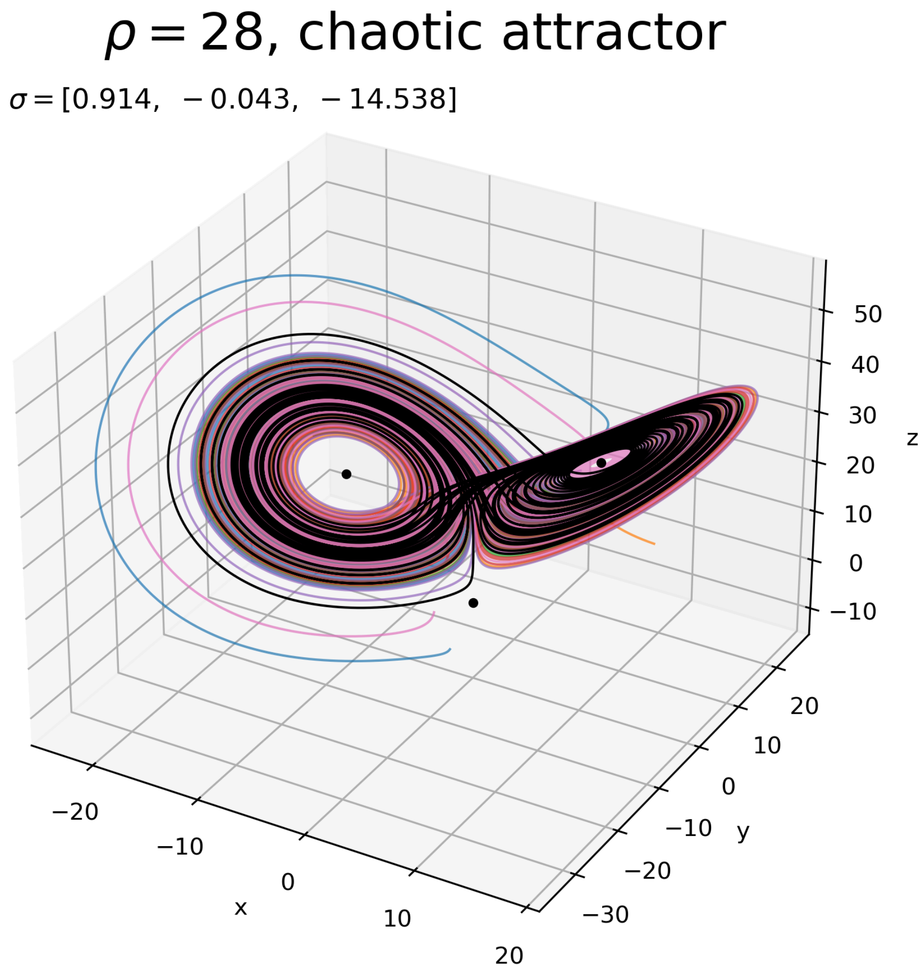
The i-th Lyapunov exponent:

$$\sigma_i = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\boldsymbol{\delta}_i(t)\|}{\|\boldsymbol{\delta}_i(0)\|}$$

- $\sigma_{\max} < 0$: all perturbations decay, **stable fixed point**
- $\sigma_{\max} = 0$: there is neutral direction (e.g. along trajectory in a limit cycle)
- $\sigma_{\max} > 0$: **sensitive dependence on initial conditions = chaos**

Lorenz-63: $\sigma_1 \approx 0.91,\ \sigma_2 = 0,\ \sigma_3 \approx -14.57$

Nonlinear dynamical system (Lorenz-63)



Dynamical systems meets data

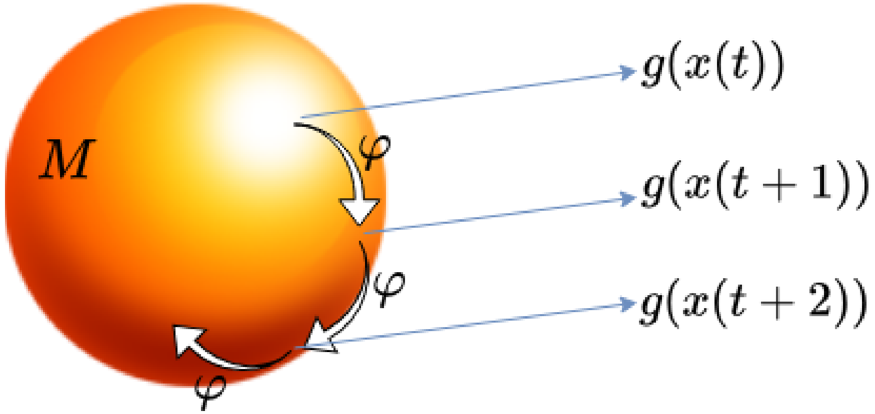
Dynamical systems meets data

So far: dynamical system is defined on a state space M, but in practice **we do not know** M or φ, they are latent (hidden).

We only observe some variables:

$$y(t) = g(\mathbf{x}(t)), \qquad g : M \to \mathbb{R}^k$$

which may or may not represent the true state, e.g. we measure temperature, but the governing equations are for pressure, velocity, density, ...



- Can we **reconstruct** the dynamics from observations alone?
- Can we find a **linear representation** of nonlinear dynamics?

State space reconstruction: Whitney embedding theorem

Whitney embedding theorem (1944): for any smooth d-dimensional manifold M, a **generic** smooth map $g : M \to \mathbb{R}^{2d+1}$ is an embedding

- "Generic" means: almost any smooth function works, non-generic cases form a set of measure zero
- g is a diffeomorphism between M and its image $g(M)$, the dynamics on M and on $g(M) \subset \mathbb{R}^{2d+1}$ are smoothly conjugate, i.e. they represent the same dynamical system in different coordinates

State space reconstruction: Whitney embedding theorem

Whitney embedding theorem (1944): for any smooth d-dimensional manifold M, a **generic** smooth map $g : M \to \mathbb{R}^{2d+1}$ is an embedding (diffeomorphism)

- "Generic" means: almost any smooth function works, non-generic cases form a set of measure zero
- g is a diffeomorphism between M and its image $g(M)$, the dynamics on M and on $g(M) \subset \mathbb{R}^{2d+1}$ are smoothly conjugate, i.e. they represent the same dynamical system in different coordinates
- In practice we do not know the dimension d, determinism or stationarity properties may not hold...
- But still it gives an intuition why $2d+1$ generic measurements (e.g. temperatures at different locations) could be sufficient to represent climate dynamics

State space reconstruction: Takens embedding theorem

Takens embedding theorem (1981): for a generic smooth system φ_τ on a $\leq d$-dimensional attractor $\mathcal{A}$ and a generic scalar observation $g : M \to \mathbb{R}$, the **delay coordinate**:

$$\mathbf{y}(t) = (g(\mathbf{x}(t)),\ g(\varphi_\tau(\mathbf{x}(t))),\ g(\varphi_{2\tau}(\mathbf{x}(t))),\ \ldots,\ g(\varphi_{2d\tau}(\mathbf{x}(t))))$$

is an **embedding** (diffeomorphism) of $\mathcal{A}$ in $\mathbb{R}^{2d+1}$ for any τ

- Practically, dynamics can be reconstructed from a **single scalar time series** $g(\mathbf{x}(t_0)), g(\mathbf{x}(t_1)), \ldots$

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- Practically, dynamics can be reconstructed from a **single scalar time series** $g(\mathbf{x}(t_0)), g(\mathbf{x}(t_1)), \ldots$
- In contrast, $2d+1$ functions of a single value $g(\mathbf{x})$ are not generic in Whitney's sense, they define a 1D curve in $\mathbb{R}^{2d+1}$, not an embedding of $\mathcal{A}$

Koopman Operator: linear dynamical system for observables

Consider a linear space G of observable functions $g(\mathbf{x})$, $\mathbf{x} \in M$, e.g. $g : M \to \mathbb{C}$

For a dynamical system $\mathbf{x}(t+\tau) = \varphi_\tau(\mathbf{x}(t))$, the **Koopman operator** $\mathcal{K}^\tau$ acts on **observable functions** $g \in G$:

$$\mathcal{K}^\tau g = g \circ \varphi_\tau, \qquad \text{i.e.} \qquad (\mathcal{K}^\tau g)(\mathbf{x}) = g(\varphi_\tau(\mathbf{x}))$$

- **If G is closed under $\mathcal{K}^\tau$**, i.e. $(K^\tau g) \in G$, then $\mathcal{K}^\tau$ is an evolution operator for observable functions: $g(\mathbf{x}(t+\tau)) = (\mathcal{K}^\tau g)(\mathbf{x}(t))$
- $\mathcal{K}^\tau$ is **linear**: $\mathcal{K}^\tau(\alpha g + \beta f) = \alpha \mathcal{K}^\tau g + \beta \mathcal{K}^\tau f$
- Acts on an **infinite-dimensional** space

Koopman Operator: linear dynamical system for observables

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- $\mathcal{K}^\tau$ is **linear**: $\mathcal{K}^\tau(\alpha g + \beta f) = \alpha \mathcal{K}^\tau g + \beta \mathcal{K}^\tau f$
- Acts on an **infinite-dimensional** space
- If G is a space with inner product (Hilbert space), then we can define a complete basis $\{\psi_k\}$, decomposition along the basis $g = \sum_k \langle g, \psi_k \rangle \psi_k$, eigenvalues, eigenmodes …

Koopman operator: linear representation of chaos?

- Nonlinear dynamical systems can have chaotic attractors
- Linear dynamical systems do not
- Koopman theory claims nonlinear dynamics can be represented linearly

Paradox?

Koopman operator: linear representation of chaos?

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So where did the chaos go?

- **Into an infinite-dimensional space of observables (functions)**
- The dynamics become linear, but the spectrum of a linear operator on a function space can be far richer than that of a finite-dimensional matrix. Welcome to functional analysis!
- In practice, finite Koopman approximation can only predict in finite time
- github.com/IPL-UV/dynamical_systems_course

Take-home messages

- Dynamical systems provide a universal language connecting mathematics, physics, observations, and models
- Linear systems are simple because eigenvalues tell the whole story
- Nonlinear systems can produce attractors, bifurcations, and chaos
- Chaos limits predictability but does not imply randomness
- Observations can contain enough information to reconstruct hidden dynamics
- Koopman theory reveals a surprising fact: nonlinear dynamics can be represented by a linear operator

- *Next*: DMD, Koopman mode decomposition, reduced representations